

# Algebra 7

## Review

$$1 \quad y' = 5x^4 - 20x^3 + 15 = 0$$

$$x^4 - 4x^3 + 3 = 0$$

$$y'' = 4x^3 - 12x^2$$

$$x^2(x-3) = 0$$

$$x=0 \quad (3)$$

$$x=3 \quad (1)$$

2 stretch by scale factor  $\frac{1}{2}$

translate by  $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$

3 log y against x

$$m = \log n$$

$$n = 10^m$$

$$c = \log k$$

$$k = 10^c$$

$$4 \quad \begin{aligned} v_1 &= 60 & = a \\ v_{25} &= -12 & -12 = 60 + (24)d \\ d &= \frac{72}{24} = 3 \end{aligned}$$

$$S_{25} = \frac{25}{2} (120 + 24 \times 3)$$

$$= 2400$$

$$5 \quad y = \frac{1}{1+\cos x} = (1+\cos x)^{-1}$$

$$y' = -(1+\cos x)^{-2} \times -\sin x$$

$$= \frac{\sin x}{1+\cos x}$$

$$At \quad x = \frac{\pi}{2}, \quad y' = \frac{1}{1+0} = 1$$

## Consolidation

$$1 \quad \frac{x^2 - 5x}{x-5} = x$$

$$\frac{x^2 + 6x - 16}{(x-2)(x-8)} = \frac{x+2}{x-2}$$

$$\frac{x^2 - a^2}{(x-a)^2} = \frac{x+a}{x-a}$$

$$\frac{v-3}{6-2v} = -\frac{1}{2}$$

$$2 \frac{5c+15}{2} \times \frac{4}{c^2-9} = \frac{20c+60}{2c^2-18}$$

$$= \frac{20}{2c-6} = \frac{10}{c-3}$$

$$3) \frac{x}{2} + \frac{3x}{5} = \frac{6x+5x}{10} = \frac{11x}{10}$$

$$ii) \frac{2}{3x} - \frac{1}{4x} = \frac{5}{12x}$$

$$4) \quad x+2 \sqrt{\frac{3x^2-7x+6}{3x^3-x^2+2x-4}} \quad r-36 \quad \frac{3x^2-7x+6}{x+2}$$

$$ii) \quad x+2 \sqrt{\frac{x}{x^3}} \quad r-2x \quad \frac{x - \frac{2x}{x^2+2}}{x^2+2}$$

$$iii) \quad 3 \quad r-11 \quad \frac{3 - \frac{11}{2x+3}}{2x+3}$$

$$5) \frac{x+7}{(x+1)(x-2)} = \frac{A}{x+1} + \frac{B}{x-2}$$

$$A(x-2) + B(x+1) = x+7$$

$$B=3$$

$$A=-2 \quad \frac{3}{x+1} - \frac{2}{x-2} = \frac{x+7}{(x+1)(x-2)}$$

$$ii) \frac{2x+1}{x^2-4} = \frac{A}{x-2} + \frac{B}{x+2}$$

$$A(x+2) + B(x-2) = 2x+1$$

$$A = \frac{3}{4}$$

$$B = \frac{1}{4}$$

$$\frac{3}{4(x-2)} + \frac{1}{4(x+2)} = \frac{x+1}{x^2-4}$$

$$6 \frac{4}{(1+3x)(1+x)^2} = \frac{A}{1+3x} + \frac{B}{1+x} + \frac{C}{(1+x)^2}$$

$$A(1+x)^2 + B(1+3x)(1+x) + C(1+3x) = 4$$

$$-2C = 4$$

$$C = -2$$

$$A = 9$$

$$A = 9$$

$$Ax^2 = 3Bx^2$$

$$3B = 9$$

$$B = 3$$

$$ii) \frac{13-7x}{(x-2)(x-3)} = \frac{A}{x-2} + \frac{B}{x-3} + \frac{C}{x-3}$$

$$A(x-3)(x-3) + B(x-3) + C(x-2)^2 = 13-7x$$

$$C = -8$$

$$B = 1$$

$$A = 8$$

$$Ax^2 - Cx^2 = 0$$

### Exam Qs

$$9) \frac{1+11x-6x^2}{(x-3)(1-2x)} = \frac{B}{x-3} + \frac{C}{1-2x} + A$$

$$A(x-3)(1-2x) + B(-2x+1) + C(x-3) = -6x^2 + 11x + 1$$

$$A = 3$$

$$A \cdot x^2 = -6x^2$$

$$B = 4$$

$$-5B = -20$$

$$C = -2$$

$$-\frac{5}{2}C = 5$$

$$6) f(x) = 3 + \frac{4}{x-3} - \frac{2}{1-2x}$$

$$(f \text{ decreasing}), f'(x) < 0 \quad (x > 3)$$

$$f(x) = 3 + 4(x-3)^{-1} - 2(1-2x)^{-1}$$

$$f'(x) = -4(x-3)^{-2} - 4(1-2x)^{-2}$$

$$= -\frac{4}{(x-3)^2} - \frac{4}{(1-2x)^2}$$

$$\text{As both denominators are } > 0 \quad (x > 3)$$

$$f'(x) < 0 \text{ for } \forall x > 3 \quad \square$$